A11y Foundations

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Foundations

- History of accessibility
- Statistics & numbers
- Types of disabilities
- Legal basis (DE, AT, CH)

Agenda

History of A11y

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History – Early 1990s

– Tim Berners-Lee, the inventor of the World Wide Web envisioned it as a platform for universal access regardless of physical or cognitive abilities

Emergence Of Accessibility – mid 90s

- People with disabilities including those with
  - visual impairments
  - hearing impairments
  - motor disabilities
  - cognitive disabilities
  - …

found it difficult or impossible to access much of the web

Web Accessibility Initiative (WAI) – 1997

– World Wide Web Consortium (W3C) establishes the Web Accessibility Initiative to address accessibility issues systematically

– WAI develops guidelines and resources to help make the web more accessible to people with disabilities

Web Content A11y Guidelines (WCAG) – 1999

– by WAI of W3C

– provides a set of recommendations for making web content more accessible to various disabilities

– 1st version published in 1999

– Version 2.1 – June, 2018 (often-used base for laws)

– **Current** Version 2.2 – October, 2023

Legal & Regulatory Development – 2000s

– Several countries started to enact laws and regulations **mandating accessibility for public sector websites**

and, in some cases, for **private sector websites** as well

Assistive Technologies – 2000s

- Simultaneously with the development of WCAG, advancements in **assistive technologies** such as
  - screen readers, Braille
  - screen magnifiers
  - alternative input devices
  - captioning tools
  - …

improved A11y for users with disabilities

Evolution of the web – 2010s

– JavaScript like **AJAX, SPAs and frameworks** like Angular emerged, posing new challenges but also offering new opportunities

– multimedia, responsive design, mobile apps required ongoing efforts to ensure A11y

Continued Advocacy & Education – 2020s

– A11y **remains an ongoing effort**, with advocacy groups, organizations, developers, designers, and policymakers continuing to work towards making the web more inclusive

Current State – 2024

– A11y has gained significant **attention and importance**, both from **a legal compliance standpoint** & **as a moral imperative**

– Organizations **increasingly recognize the business benefits of accessible design**, including expanded reach, improved user experience for all users, and compliance with legal requirements

Some statistics

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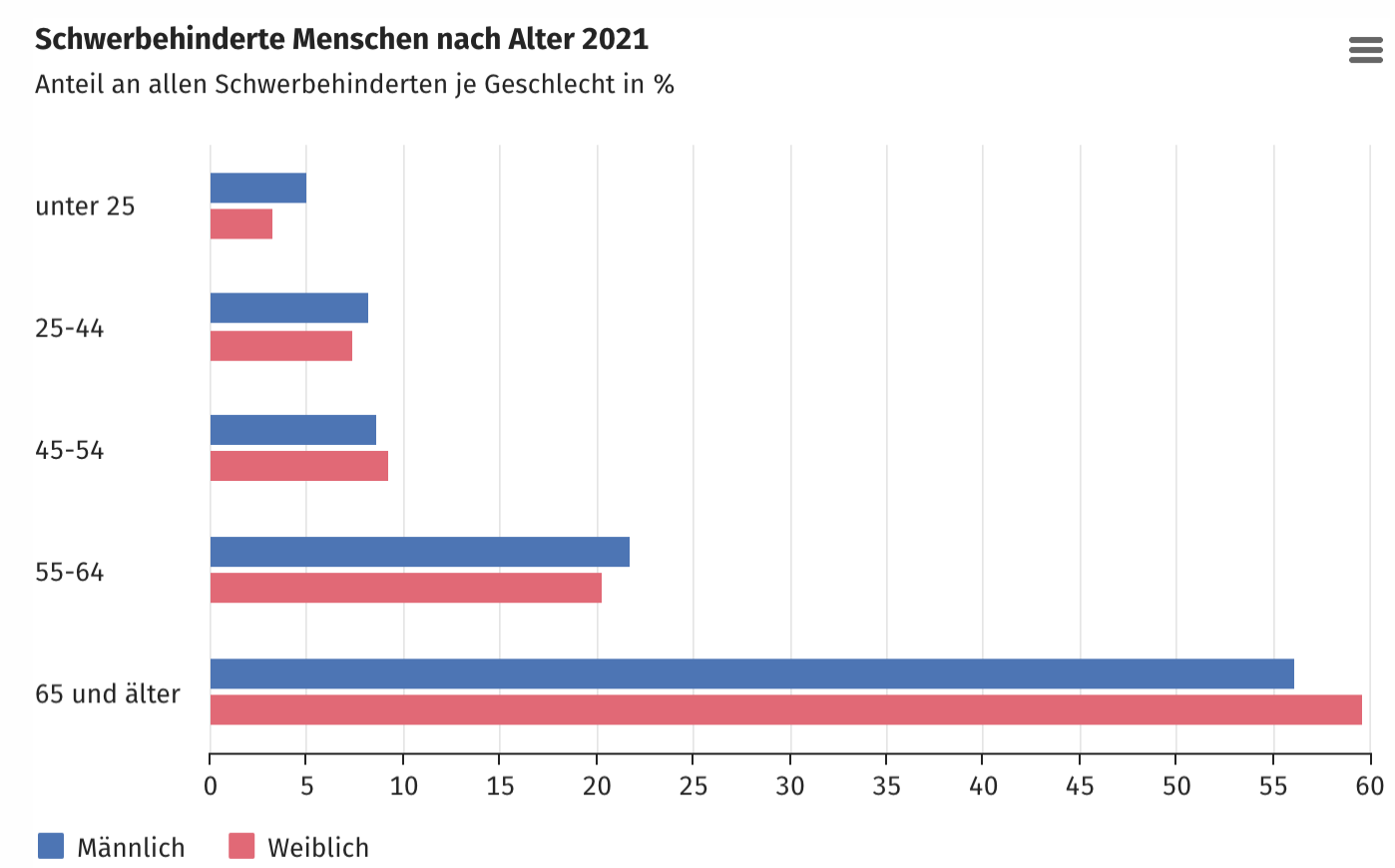
Germany

- 7,8 Mio. (9,4%) severely disabled
- > 10 Mio. (12,7%) officially recognized disab.
- (Statistisches Bundesamt, 2021)
- 18,7 Mio (22 %) aged >= 65 (2022)

Numbers

Germany



Personen mit Signierung des Geschlechts "divers" und "ohne Angabe" (§ 22 Absatz 3 PStG) werden ab dem Berichtsjahr 2021 aus Gründen der statistischen Geheimhaltung per Zufallsprinzip dem männlichen oder weiblichen Geschlecht zugeordnet.

Numbers

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Austria

- 1,34 Mio. (18,4%) permanently disabled
  - 1 Mio. (14,1%) problems with mobility
  - 216k (3,0%) problems with vision
  - 157k (2,1%) problems with hearing
  - 60k (0,8%) mental illness / learning disabilities

- (Statistik Austria, 2015)

Numbers

Types of disabilities

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Classic disabilities

- Mobility (14,1% AT)
- Vision (3,0% AT)
- Hearing (2,1% AT)
- Mental / learning disabilities (0,8% AT)

Other disabilities

- Dementia (2% DE)
- Non-native speakers (20% DE / 24% AT)
- Reading/spelling weakness (12% DE / 10% AT)
- Parents (just 1 hand ☺)

Legal basis

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Behindertengleichstellungsgesetz (BGG)

Disability Equality Act

– DE – May 1st 2002

– prevent discrimination against people with disabilities

– ensure the equal participation of people with disabilities in life in society

– enable them to lead a self-determined lifestyle

Legal

Barrierefreie-Informationstechnik-Verordnung (BITV 2.0)

Barrier-free Information Technology Regulation

- DE V1.0 2002 based upon WCAG 1.0 by WAI
- All public web services must be accessible
  - since December 31st 2005
- 2011 (BITV 2.0) & 2019 updates – WCAG 2.0
  - since March 22nd 2012

Legal

Barrierefreiheitserklärung

Accessibility Statement

- own page (like imprint / data protection)
  - link placed in footer
- machine readable
- mention national law
- name all contents that aren't yet accessible
- examples
  - https://www.bundesfinanzministerium.de/Web/DE/Meta/Erklaerung-Barrierefreiheit/erklaerung-barrierefreiheit.html
  - https://www.bmf.gv.at/public/barrierefreiheitserklaerung.html

Legal

European Accessibility Act 2025 (EAA)

- EU directive from April 2019 (!)
  - Europäisches Gesetz zur Barrierefreiheit (EGB)

- regulations for the accessible use of certain products and services such as
  - terminals
    - ticket machines in public transport systems
    - payment terminals and cash machines
  - products
    - computers
    - mobile phones
  - services
    - e-commerce (online shops, everything that creates contracts)
    - e-books

Legal

Barrierefreiheitsstärkungsgesetz (BFSG)

Accessibility Strengthening Act

- DE – July 2022 national application of the EAA
- All private web services must be accessible too
  - Starting from June 28th 2025 all new products & services
  - immediately applicable for online services like shops
  - up to 5 years for migration of existing products
  - up to 15 years for migration of ticket machines / terminals
- Includes online services like banking & shops
- Small bus. (<10 emp. or <2 M. € sales) excluded
- https://www.digitalmanufaktur.com/blog/barrierefreiheitsstarkungsgesetz-2025-was-onlineshops-jetzt-tun-mussen

Legal

Barrierefreiheitsgesetz (BaFG)

Accessibility Act

- AT – June 2023 national application of the EAA
- All private web services must be accessible too
  - Starting from June 28th 2025 all new products & services
  - up to 5 years for migration of existing services & products
  - up to 20 years for migration of ticket machines / terminals
- Includes online services like banking & shops
- Small bus. (<10 emp. or <2 M. € sales) excluded
- Ref: https://www.wko.at/ce-kennzeichnung-normen/informationen-zum-barrierefreiheitsgesetz

Legal

A11y Standard (eCH-0059)

- CH
  - Version 1.0 – Aug. 2007
  - Version 2.0 – Feb. 2011
  - Version 3.0 – **June 2020**

- Current version
  - Inspired by EU's EAA
  - Also based upon **WCAG 2.1**

Legal

Questions?

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